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DATA PIPELINE AND METRIC GENERATION

In various embodiments, the computing system implements a data pipeline configured to transform raw signals from one or more nodes into features, metrics, and privacy-preserving outputs. The pipeline may operate in real time, near real time, or in batch mode, and may be distributed across node firmware, a mobile device, and/or a cloud service.

A. Data Acquisition and Synchronization

Each node may acquire one or more raw signals at one or more sampling rates. Sampling rates may be fixed or adaptive. In certain embodiments, the computing system synchronizes data streams from multiple nodes using timestamps, beacon packets, clock correction, event markers, and/or alignment routines. In some embodiments, the system performs time-window segmentation comprising fixed windows, rolling windows, or event-triggered windows.

B. Pre-Processing and Signal Conditioning

The computing system may perform signal conditioning including filtering, baseline removal, drift compensation, normalization, temperature compensation, motion gating, and other transforms. In certain embodiments, the system computes one or more signal quality indices based on contact integrity, sensor saturation, motion state, and/or multi-sensor agreement.

C. Artifact Rejection and Wear Validation

In various embodiments, the pipeline includes artifact rejection to reduce distortion from motion, misalignment, poor contact, or environmental changes. Artifact rejection may be performed using inertial sensing, contact integrity sensing, capacitive proximity/contact confirmation, impedance consistency checks, optical coupling checks, and combinations thereof. In certain embodiments, the system performs wear validation to determine whether a node is being worn in a valid context, and excludes invalid segments from analysis.

D. Feature Extraction

The computing system may derive features from conditioned signals, including peaks, slopes, durations, rise and fall times, area-under-curve values, variability measures, frequency-domain measures, periodicity measures, stability measures, and multi-sensor consistency measures. In some embodiments, features are derived per node and per modality, and additional features are derived from relationships between modalities.

E. Metric Generation and Privacy-Preserving Outputs

The computing system may transform features into metrics configured to represent physiologic patterns without requiring explicit anatomical imagery. Metrics may include indices, ratios, zones, ranges, trend measures, temporal signatures, variability signatures, and confidence indicators. Outputs may be presented using abstract bars, curves, zones, and comparative time-series views. In certain embodiments, the system presents results as within-user change over time and/or as correlations between node outputs.

F. Cross-Device Correlation

In various embodiments, the computing system performs cross-device correlation between outputs derived from different nodes. Cross-device correlation may include temporal alignment, correlation scoring, lag analysis, sequence detection, co-variation analysis, contextual conditioning, and confidence scoring. For example, intimate-region metrics may be correlated with cardiophysiologic context metrics, activity state, temperature state, baseline acquisition results, or other contextual parameters to improve interpretation and reduce ambiguity.

G. Cross-User Correlation

In certain embodiments, the computing system performs privacy-preserving cross-user correlation, in which a first user’s metrics are compared with a second user’s metrics without requiring anatomical imaging or explicit anatomical labeling. Cross-user correlation may be performed across a range of pairing contexts including male-to-female, male-to-male, female-to-female, and other pairings. In some embodiments, the system computes correlation scores, alignment scores, stability scores, and/or timing synchrony measures based on abstracted metrics.